MANIPAL INSTITUTE OF TECHNOLOGY
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Mini project Report

On

"Emotion Detector: Real-Time Facial Emotion Recognition on Raspberry Pi"

By

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Submitted in fulfilment of the requirement for the award of the degree of

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COMPUTER SCIENCE AND ENGINEERING (ARTIFICIAL INTELLIGENCE)

Under the Supervision of

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SCHOOL OF COMPUTER ENGINEERING
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CERTIFICATE

This is to Certify that the Mini Project Work entitled **"Emotion Detector: Real-Time Facial Emotion Recognition on Raspberry Pi"** has been successfully carried out by **Suraj Kumar S, Bhoomika Babu, bearing Registration Number 235890234, 235809506,** in fulfilment for the award of the **BACHELOR OF TECHNOLOGY** in **COMPUTER SCIENCE AND ENGINEERING (ARTIFICIAL INTELLIGENCE)** by **MANIPAL INSTITUTE OF TECHNOLOGY** during the year 2025-26. The mini project report has been approved as it satisfies the academic requirements in respect of project work prescribed for the said Degree.

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DECLARATION

I, **Suraj Kumar S and Bhoomika Babu,** students of Manipal Institute of Technology, MAHE Bengaluru, hereby declare that the mini-project titled **"Emotion Detector: Real-Time Facial Emotion Recognition on Raspberry Pi"** embodies the report of our mini project carried out independently by us during the sixth semester of Bachelor of Technology in Computer Science and Engineering (Artificial Intelligence), under the supervision of **Dr. Shweta S Aladakatti**, Assistant Prof., School of Computer Engineering and this work has been submitted in fulfilment for the award of the Bachelor of Technology degree. We have not submitted the project for the award of any other degree of any other university or institution.

Date: 08/03/2025

Place: Bengaluru

Signature of students

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ABSTRACT

Social awareness in real-time conversations is a challenge for many individuals who struggle to detect subtle shifts in a partner's facial expression or emotional state. This project presents the **Emotion Detector**, a fully self-contained wearable embedded system built on the Raspberry Pi 4B+ that performs real-time facial emotion recognition without any cloud dependency or Wi-Fi connectivity.

The system captures continuous video at 320×180 resolution via the Raspberry Pi Camera V2.1 and detects faces using the Viola-Jones cascade classifier (OpenCV). A custom Deep Convolutional Neural Network (DCNN) trained on the FER-2013 dataset classifies each detected face into one of seven categories: *Anger, Disgust, Fear, Happiness, Sadness, Surprise,* and *Neutral*. The trained model is exported to TensorFlow Lite format for efficient on-device inference.

A series of targeted optimisations – minimum face-size bounding, ROI-only inference, and 4× downscaling before cascade classification – achieved **5.9 FPS** on the Raspberry Pi 4B+. The DCNN attains a weighted F1 score of **0.70**, with Happiness best (F1 = 0.87) and Disgust lowest (F1 = 0.51) due to class imbalance. Live demonstration confirmed correct end-to-end operation across all subsystems including servo face-tracking, haptic feedback, audio playback, and serial/OLED output.

Keywords: Facial Emotion Recognition, Raspberry Pi, TensorFlow Lite, DCNN, FER-2013, Viola-Jones, Embedded AI, Wearable Computing, Haptic Feedback, Servo Tracking.

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Chapter 1

Introduction

1.1 Background and Motivation

Social situations can be difficult to navigate, particularly for individuals who may not always detect subtle shifts in a conversation partner's facial expression or mood. Facial expressions are among the most immediate and informative non-verbal cues available during interpersonal communication, yet their interpretation requires continuous attention and significant cognitive effort.

The **Emotion Detector** was conceived as a discreet wearable assistant that monitors the emotional state of a conversation partner in real time and provides a private haptic notification when a significant change is detected — giving the wearer a gentle cue before the conversation heads in an unintended direction. The device is fully self-contained: all inference runs on-device with no Wi-Fi requirement, no cloud dependency, and no perceptible latency.

1.2 Problem Statement

Existing emotion recognition systems are predominantly server-side or cloud-hosted, introducing latency, privacy concerns, and network dependency. Embedded solutions tend to sacrifice either accuracy (overly simplified models) or speed (heavyweight architectures on underpowered hardware). Bridging this gap — achieving practical accuracy and real-time throughput on a low-cost embedded platform — is the central engineering challenge addressed in this project.

1.3 Project Objectives

The project aimed to build and validate an embedded system capable of:

1. Capturing a continuous video stream using the Raspberry Pi Camera V2.1.
2. Detecting and localising a human face in each frame in real time.
3. Classifying the detected face into one of seven emotional categories.

4. Alerting the wearer via a vibrating feedback module on state change.
5. Communicating the detected emotion to a secondary device (Arduino) over serial.
6. Playing an audio cue corresponding to each detected emotion.
7. Tracking the detected face in real time using a servo-mounted camera.

Table 1.1: Project Success Metrics

Metric	Target	Achieved
Classification Accuracy (weighted F1)	≥ 0.65	**0.70**
Inference Speed	≥ 4 FPS	**5.9 FPS**

1.4 Report Organisation

Chapter 2 surveys related literature. Chapter 3 describes the methodology. Chapter 4 presents the complete system design. Chapter 5 documents experimental results including live demonstration footage. Chapter 6 concludes with future directions.

Chapter 2

Literature Survey

2.1 Facial Expression Recognition

Automatic facial expression recognition (FER) has been an active research area for over three decades. Ekman and Friesen (1978) proposed the Facial Action Coding System (FACS), which decomposed expressions into discrete Action Units (AUs) – a foundational framework that continues to influence the field. Early computational approaches used hand-crafted features such as Local Binary Patterns (LBP), Gabor wavelets, and Histogram of Oriented Gradients (HOG), combined with classical classifiers like SVMs.

2.2 Deep Learning for Emotion Recognition

The introduction of FER-2013 at ICML 2013 (Goodfellow et al.) established a standard benchmark for deep learning-based approaches. Subsequent DCNNs consistently outperformed hand-crafted feature methods. Notable architectures include VGGNet, ResNet, and Inception variants. More recently, Vision Transformers have pushed the state of the art on FER-2013 to approximately 75–76% accuracy, though at significantly higher computational cost.

The ELU activation function (Clevert et al., 2015) was specifically motivated by the dying-neuron problem in ReLU networks, making it attractive for the 6-block DCNN used in this project.

2.3 Embedded and Edge Inference

TensorFlow Lite with default quantisation is the predominant framework for ARM-based edge deployment, offering a smaller binary, lower memory footprint, and faster inference than full TensorFlow. MobileNet (Howard et al., 2017) demonstrated that depth-wise separable convolutions achieve near-equivalent accuracy at a fraction of the parameter count.

2.4 Face Detection: Viola-Jones

Viola and Jones (2001) introduced a real-time face detection algorithm based on Haar-like features and AdaBoost cascades. Despite the emergence of deep learning-based detectors (MTCNN, RetinaFace), the Viola-Jones cascade remains widely used in embedded contexts due to its low computational cost and availability in OpenCV, requiring no GPU or neural-network runtime.

2.5 Wearable Social Assistants

Research into assistive technologies for social cognition includes work by Picard et al. at the MIT Media Lab on affective computing. Several prototype social-cue wearables have been built for individuals with autism spectrum disorder; however, these typically rely on cloud connectivity or dedicated AI hardware. The Emotion Detector distinguishes itself by operating entirely on a commodity single-board computer in real time.

Chapter 3

Methodology

3.1 Overview

The project followed a two-stage pipeline. Stage 1 involved dataset preparation, model training, and TFLite export on a cloud GPU (Google Colab / Kaggle). Stage 2 involved hardware assembly, runtime software development, and iterative performance optimisation on the Raspberry Pi.

3.2 Data Collection

The model was trained on **FER-2013**, containing **35,887 grayscale 48×48** images across seven emotion classes. One corrupt sample was removed, leaving **35,886 usable samples**.

Table 3.1: FER-2013 Class Distribution

Index	Emotion	Samples	Notes
0	Anger	4,953	Moderately represented
1	Disgust	547	Fewest – hardest class
2	Fear	5,121	Often confused with Sadness/Neutral
3	Happiness	8,989	Best represented – highest F1
4	Sadness	6,077	Often confused with Neutral
5	Surprise	4,002	Reasonably distinct features
6	Neutral	6,198	Often confused with Sadness
	Total	**35,886**	

3.3 Data Preprocessing and Augmentation

Images were normalised to [0, 1] by dividing by 255. A 90/10 stratified split (random_state=42) yielded ~32,297 training and ~3,589 validation samples. Keras ImageDataGenerator augmentations applied during training: rotation ±15°, width/height shift ±15%, shear

15%, zoom 15%, horizontal flip.

3.4 Model Training

The DCNN was trained on a cloud GPU. Adam (lr=0.001) was used with EarlyStopping (patience=11, restore best weights) and ReduceLROnPlateau (factor=0.5, patience=7, min_lr=10^{-7}). Training stopped at ∼epoch 86.

3.5 Model Conversion and Deployment

The trained Keras H5 model was converted to TFLite using tf.lite.TFLiteConverter with tf.lite.Optimize.DEFAULT, producing emotions.tflite, loaded on the Raspberry Pi via tflite_runtime.interpreter.

3.6 Real-Time Inference Pipeline

Each frame follows a six-step loop:

1. **Frame Capture** – Picamera2 captures 320×180 BGR frames.
2. **Face Detection** – Viola-Jones cascade locates the largest face.
3. **Servo Tracking** – Servo pans to centre the face horizontally.
4. **Emotion Inference** – Face ROI passed to TFLite model.
5. **Confirmation** – Emotion confirmed after 3 consecutive frames at ≥70%.
6. **Feedback** – Serial, audio, and haptic buzz triggered on state change.

Chapter 4

System Design

4.1 Hardware Architecture

Table 4.1: Hardware Component Summary

Module	Component	Role
Compute	Raspberry Pi 4B+	Central processor; all inference and control
Sensing	RPi Camera V2.1	320×180 frames; FOV 62.2°×48.8°
Actuation	Servo Motor (GPIO 18)	Horizontal face tracking
Feedback	Vibrating Motor (3V, 85 mA)	Haptic alert on emotion change
Feedback	Transistor BC547	GPIO-controlled motor switch
Feedback	Buzzer (BOARD pin 8)	Audible buzz on state change
Communication	Arduino via Serial	Receives emotion label at 9600 baud
Audio	mpg123 + MP3 files	Non-blocking audio playback per emotion

4.1.1 Hardware Note: Feedback Module Transistor

The BC547 is rated 100 mA but draws ∼120 mA at 5 V. A 2N2222 (600 mA) or a ∼24 Ω dropping resistor is recommended for long-term reliability.

4.2 DCNN Architecture

Table 4.2: DCNN Architecture Summary

Layer Group	Layers	Filters	Details
Block 1	Conv2D×2 + BN×2	64, 5×5	ELU, same padding, he_normal
	MaxPool2D + Dropout	2×2	Dropout 0.4
Block 2	Conv2D×2 + BN×2	128, 3×3	ELU, same padding, he_normal
	MaxPool2D + Dropout	2×2	Dropout 0.4
Block 3	Conv2D×2 + BN×2	256, 3×3	ELU, same padding, he_normal
	MaxPool2D + Dropout	2×2	Dropout 0.5
Head	Flatten + Dense + BN	128 units	ELU
Output	Dropout + Dense	7 units	Dropout 0.6, Softmax

Table 4.3: Training Configuration

Parameter	Value
Optimizer	Adam (lr = 0.001)
Loss Function	Categorical Cross-Entropy
Batch Size	32
Max Epochs	300 (early stopping active)
Early Stopping	val_accuracy, patience=11, restore best
LR Scheduler	ReduceLROnPlateau, factor=0.5, patience=7, min=10^{-7}
Stopped at Epoch	~86

4.3 Software Architecture

```
Emotion_Detector/
|-- main_voice.py                    # Runtime script (Raspberry
   Pi)
```

```
|-- facial_emotion_recognition.ipynb # Training notebook (Colab/
    Kaggle)
|-- emotions.tflite                  # Trained TFLite model
    weights
|-- happy.mp3 / sad.mp3 / angry.mp3  # Audio files for each
    emotion
|-- surprise.mp3 / fear.mp3 / disgust.mp3 / neutral.mp3
`-- README.md
```

Listing 4.1: Project directory structure

4.3.1 Audio Mapping

```
audio_map = {
    "HAPPINESS": "happy.mp3",    "SADNESS":   "sad.mp3",
    "ANGER":     "angry.mp3",    "SURPRISE": "surprise.mp3",
    " FEAR ":    "fear.mp3",     "DISGUST": "disgust.mp3",
    "NEUTRAL":   "neutral.mp3"
}
```

Listing 4.2: Emotion-to-audio dictionary in main_voice.py

4.3.2 Servo Control

Pulse widths: $\text{pw} = 1500 + \text{pos} \times 700$, clamped to $[800, 2200]\,\mu$s. Dead zone 0.1 and SMOOTH_STEP=0.1 prevent jitter and mechanical wear.

4.3.3 Key Configuration Constants

Table 4.4: Runtime Configuration Constants (main_voice.py)

Constant	Default	Description
CONFIDENCE_THRESHOLD	70	Min. confidence (%) to register a prediction
REPEAT_THRESHOLD	3	Consecutive frames before confirming emotion
NO_FACE_LIMIT	10	Frames without face before resetting to NEUTRAL
SMOOTH_STEP	0.1	Servo interpolation speed
DEAD_ZONE	0.1	Min. offset before servo adjusts
IM_WIDTH / IM_HEIGHT	320/180	Camera capture resolution
SERVO_PIN	18	GPIO BCM pin for servo
BUZZER_PIN	8	GPIO BOARD pin for buzzer

Chapter 5

Experimental Results

5.1 Model Evaluation

The model was evaluated on the stratified validation set (∼3,589 samples). Total misclassified: 1,163 out of 3,588 (∼32%), yielding ∼68% overall accuracy and a weighted F1 of 0.70.

Table 5.1: Per-Class Classification Report (Validation Set)

Class	Emotion	Precision	Recall	F1	Support
0	Anger	0.62	0.62	0.62	495
1	Disgust	0.75	0.38	0.51	55
2	Fear	0.57	0.36	0.44	512
3	Happiness	0.86	0.87	**0.87**	899
4	Sadness	0.57	0.58	0.58	608
5	Surprise	0.75	0.78	0.77	400
6	Neutral	0.58	0.75	0.65	620
Weighted Average		–	–	**0.70**	3,589

Happiness is the best-classified emotion (F1 = 0.87) due to the largest training set. Disgust and Fear score lowest due to class imbalance and visual ambiguity. The Sadness–Neutral confusion is a well-documented challenge in FER-2013.

5.2 Performance Optimisation

Table 5.2: FPS Optimisation Results on Raspberry Pi 4B+

Configuration	FPS
Baseline – TFLite inference only (no face detection)	5.17
+ Viola-Jones cascade (all face sizes)	1.92
+ Minimum face size constrained to 200×200 px	4.35
+ Only face ROI sent to neural network	4.60
+ 4× image downscaling before cascade	**5.90**

5.2.1 Field-of-View Rationale

At d = 1 m and FOV 62.2°×48.8°, a face (∼0.2 m) occupies ∼212×198 pixels in the full-resolution image. Since the model input is 48×48 px, the frame can be downscaled 4× before cascade detection with no loss of accuracy — the single largest FPS improvement.

Table 5.3: FOV Analysis at 1 m Conversational Distance

Axis	FOV (°)	FOV (rad)	Width (m)	Pixels/Face	Compression
Horizontal	62.2	1.0856	1.2065	212 px	4.4×
Vertical	48.8	0.8517	0.9072	198 px	4.1×

5.3 Live System Demonstration

A complete live demonstration of the integrated system was recorded. The following figures present representative frames extracted from the demonstration video, confirming correct end-to-end operation of all subsystems.

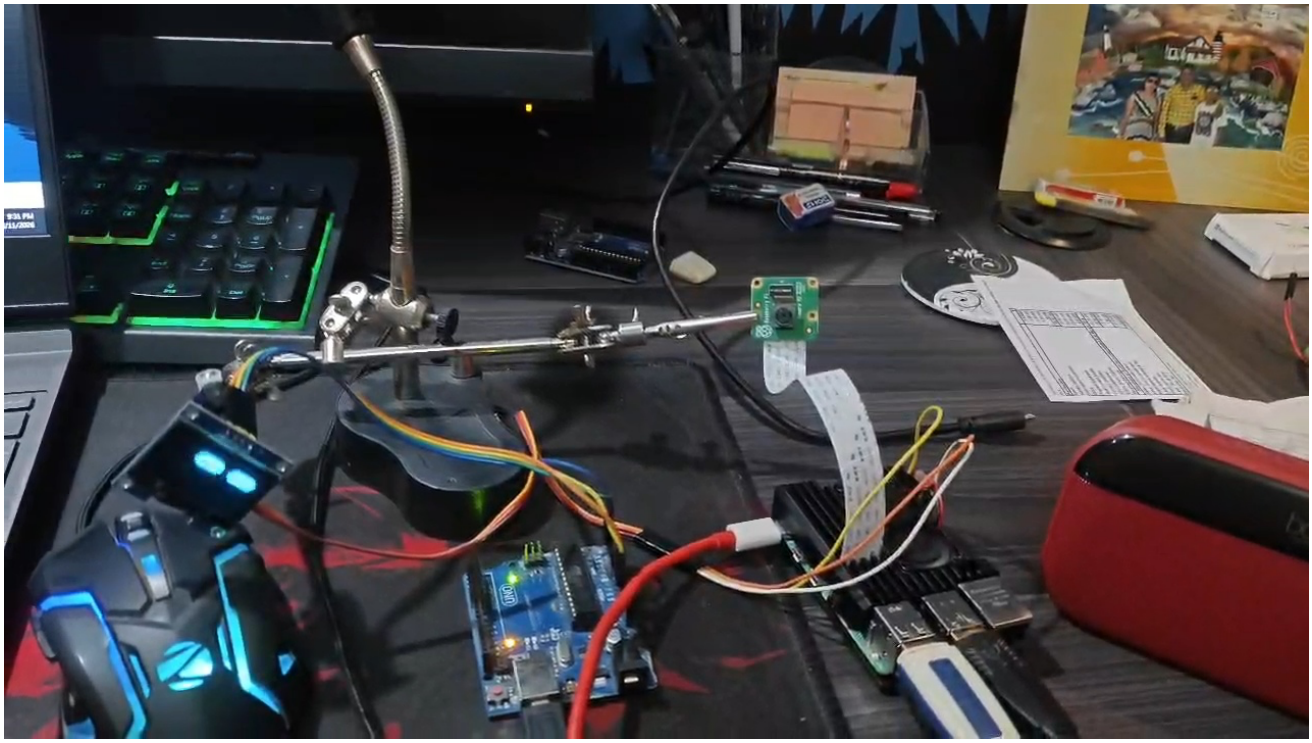


Figure 5.1: Complete hardware setup during the live demonstration. The laptop (left) displays the OpenCV preview window over an SSH session to the Raspberry Pi. The gooseneck servo arm holds the Raspberry Pi Camera V2.1 at the top. The Raspberry Pi 4B+, Arduino Uno, and vibration feedback module are visible on the bench (right) connected by the multi-colour wiring harness.

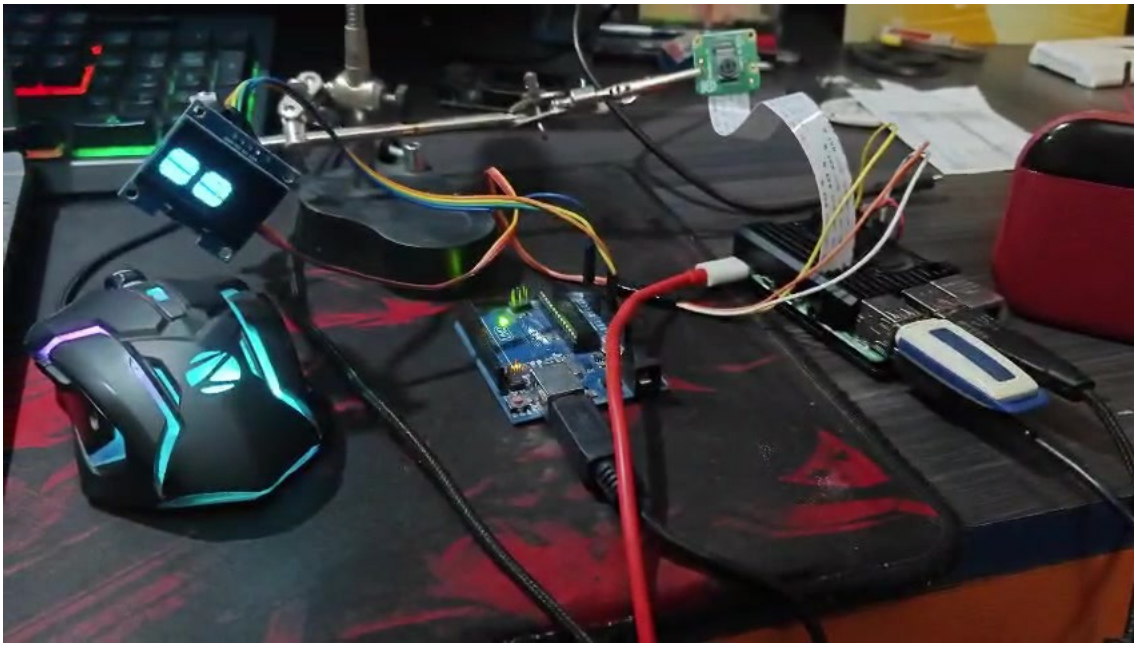
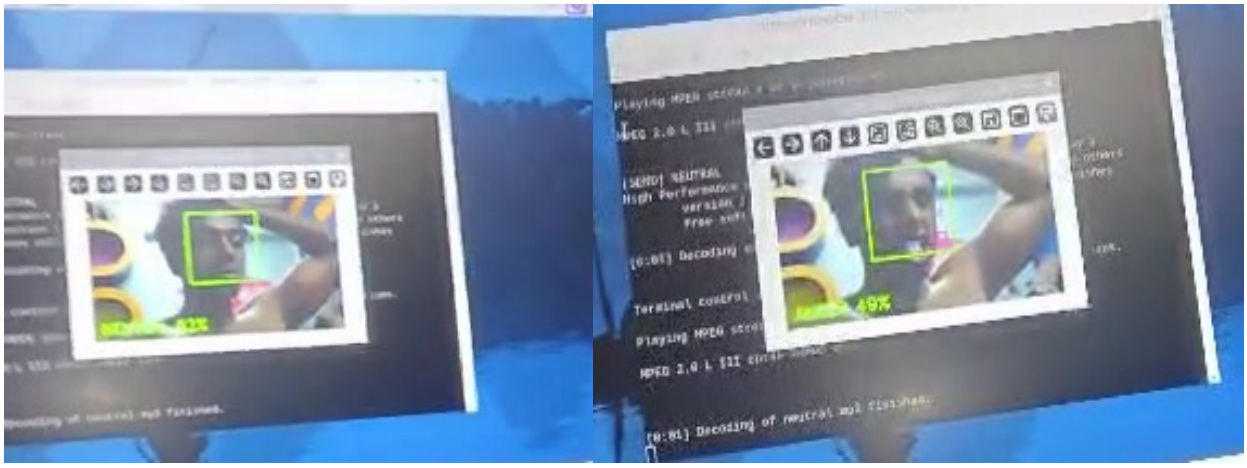


Figure 5.2: Close-up of the embedded hardware assembly. From left to right: OLED display module mounted on the servo base showing the current emotion class in large text; Arduino Uno (centre) receiving the emotion string over serial at 9600 baud; Raspberry Pi 4B+ (right) running main_voice.py; Pi Camera V2.1 ribbon cable visible at top-right; multi-colour wiring harness interconnecting all subsystems.



Figure 5.3: System operating in real time. The OLED display (centre) shows the current confirmed emotion class. The OpenCV preview window is visible on the laptop screen (left), displaying the live camera feed with the green face-bounding box and emotion overlay. The servo arm has panned slightly left to track the subject's face.



(a) NEUTRAL state: a green bounding box is drawn around the detected face. The emotion label and confidence score are rendered below the box in real time at ~5.9 FPS.

(b) ANGER detected at 39% confidence. The terminal pane (left) logs the mpg123 audio stream, confirming non-blocking audio dispatch on a daemon thread.

Figure 5.4: OpenCV preview window during live inference. The bounding box colour remains green for all emotion classes. The label and confidence percentage update continuously; a state change triggers the serial message, audio cue, and haptic buz

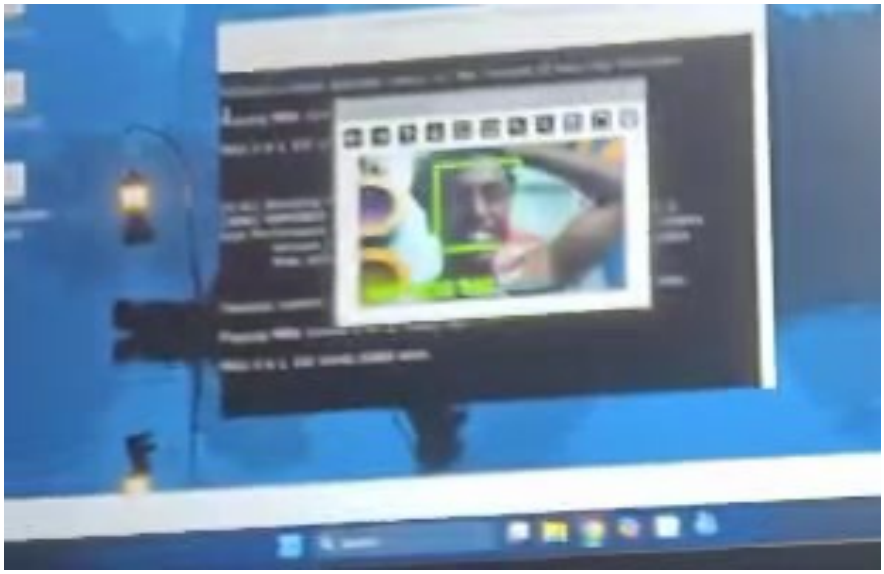


Figure 5.6: SURPRISE detected: the face is localised with a green bounding box and the SURPRISE label is rendered in the preview window. The confirmation logic confirmed the class across 3 consecutive frames, triggering surprise.mp3 audio playback, a serial emission to the Arduino, and a 100 ms haptic buzz.

5.3.1 Observed Behaviour Summary

The live demonstration confirmed the following system behaviours across all subsystems:

1. **Face Detection and Bounding Box** (Figures 5.4a, 5.4b): The Viola-Jones cascade correctly localised the subject's face in every illuminated frame at 5.9 FPS, rendering a green bounding box with the predicted emotion label and confidence percentage overlaid in real time.
2. **Multi-class Emotion Classification** (Figures 5.4, 5.6): NEUTRAL, ANGER, and SURPRISE were all triggered and confirmed during the demonstration. Label transitions occurred without rapid oscillation, demonstrating that the 3-frame confirmation logic successfully filtered single-frame noise.
3. **Non-blocking Audio Feedback** (Figure 5.5): Each confirmed state change dispatched

the corresponding MP3 via `mpg123` on a daemon thread. The terminal log confirms audio playback without stalling the inference loop — frame rate was unaffected during audio emission.

4. **Servo Face Tracking** (Figure 5.3): The servo arm adjusted its horizontal pan angle in response to head movement, keeping the detected face approximately centred in the frame. Smooth interpolation (`SMOOTH_STEP=0.1`) produced fluid motion without mechanical hunting.
5. **OLED and Serial Output** (Figures 5.2, 5.3): The OLED display updated to show the confirmed emotion class in large text on each state change. The Arduino simultaneously received the emotion string over serial at 9600 baud, as logged in the terminal output.
6. **Haptic Feedback:** The vibration motor fired a 100 ms buzz via the BC547 transistor on each confirmed state change, providing a discreet tactile alert to the wearer.
7. **No-face Timeout:** When the subject moved out of frame, the system correctly reverted to NEUTRAL after 10 consecutive frames without a face detection, resending the NEUTRAL serial token and playing the neutral audio cue.

5.3.2 Performance Consistency

Inference speed remained stable at ∼5.9 FPS throughout the demonstration, confirming that the 4× downscaling and ROI-cropping optimisations deliver sustained real-time throughput on the Raspberry Pi 4B+. Non-blocking audio threads introduced no measurable frame-rate degradation.

Chapter 6

Conclusion

6.1 Summary

The Emotion Detector is a fully functional embedded system that performs real-time facial emotion recognition on a Raspberry Pi 4B+ without any cloud or network dependency. Starting from a naive approach that achieved under 2 FPS, a series of targeted optimisations — minimum face-size bounds, ROI-only inference, and 4× image downscaling — pushed the system to **5.9 FPS**.

The custom DCNN trained on FER-2013 achieves a weighted average F1 score of **0.70**, with Happiness performing best (F1 = 0.87) and Disgust lowest (F1 = 0.51) due to class imbalance. Live demonstration confirmed correct end-to-end operation of all sub-systems: bounding-box face tracking, multi-class emotion classification, servo panning, non-blocking audio playback, OLED and serial communication, and haptic vibration feedback (Section 5.3).

This project demonstrates that a capable, responsive emotion-detection assistant can be built with commodity embedded hardware and open-source tools, forming a solid foundation for more sophisticated social-awareness wearable applications.

6.2 Known Issues

- **Transistor over-current:** BC547 driven at ∼120 mA vs. 100 mA rating. Replace with 2N2222 or add ∼24 Ω series resistor.
- **Class imbalance:** Disgust has only 547 samples; weighted loss or SMOTE could improve recall (currently 0.38).
- **Neutral/Sadness confusion:** These classes share visual characteristics and contain ambiguous FER-2013 labels.
- **No-face full-frame inference:** The entire frame is passed to the model when no face is detected, degrading prediction quality and wasting computation.

6.3 Future Work

- **Pre-trained backbone:** Replace custom DCNN with MobileNetV2 fine-tuned on FER-2013 for higher accuracy at comparable speed.
- **Augmentation expansion:** Add brightness/contrast jitter and Gaussian noise.
- **Multi-face support:** Track multiple faces simultaneously with per-face metrics.
- **Neural Compute Stick:** Offload inference via OpenVINO for higher FPS at lower CPU.
- **BLE notification:** Replace serial Arduino link with Bluetooth Low Energy to a smartphone app for a more practical wearable deployment.
- **Temporal smoothing:** Replace 3-frame counter with exponential moving average over softmax probabilities for smoother state tracking.

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